



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 7

AC Analysis I

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Topic 7 Content

This lecture covers:

- Alternating currents (AC).
 - Review of sinusoids as AC signals.
 - Review of some trigonometry properties and identities.
- Phasor representation of sinusoids.
 - Review of complex numbers and their properties.
 - Phasor relationship of electric circuit elements (resistor, capacitor, inductor).
 - Differences between time-domain and phasor/frequency-domain analysis.
- Concept of impedance and admittance.
- KCL and KVL in frequency domain.

**Corresponds to Chapter 9
of your textbook**

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO3*: Apply phasors and sinusoidal steady state analysis to AC circuits.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Alternating current

- Alternating current, or AC, is the **dominant** form of **electrical power** that is delivered to homes and industry.
- An AC current/voltage has **alternating positive** and **negative** values.
- Circuits driven by **sinusoidal** current or voltage sources are called **AC circuits**.
 - A **sinusoid** is a signal that has the form of **sine** or **cosine** function.

Sinusoids

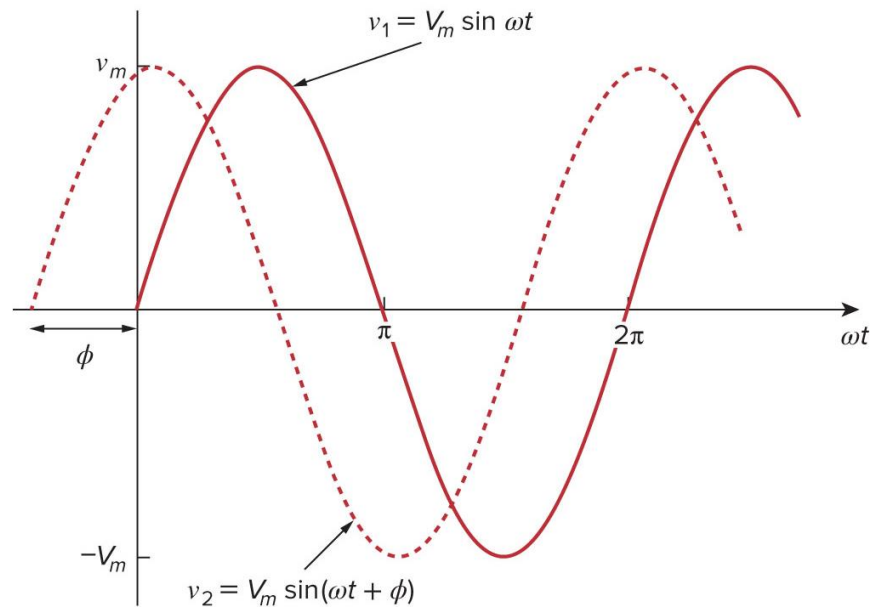
- Sinusoids can be found in many natural phenomena:
 - Pendulum motion.
 - String vibration.
 - Ocean surface ripples.
- Are easy to generate and transfer.
- Are easy to handle mathematically.

Sinusoids

- A sinusoidal voltage can be written as:

$$v(t) = V_m \sin(\omega t + \phi)$$

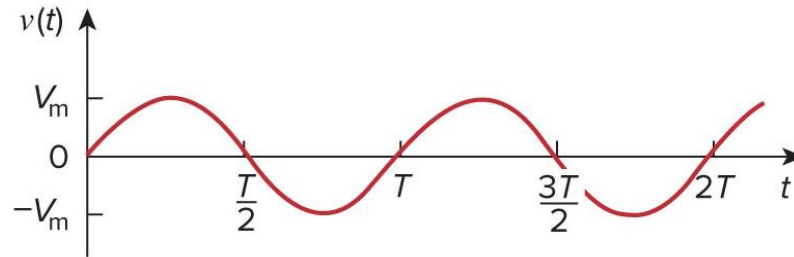
- V_m : **Amplitude** of the sinusoid.
- ω : **Angular frequency** in rad/s.
- $(\omega t + \phi)$: **Argument** of the sinusoid.
- ϕ : Phase (in degrees or radians).



Sinusoids – Period and frequency

- A **periodic function** is the one that satisfies $f(t) = f(t + nT)$ for all t and for all integers n .
- $v(t)$ repeats itself every T seconds, which is called the **period** of the sinusoid.
 - The period T is the time of **one complete cycle** or the number of seconds per cycle.
 - For a sinusoid:

$$T = \frac{2\pi}{\omega}$$



- The **reciprocal** of the period is the number of cycles per second known as the **cyclic frequency** f of the sinusoid. It is measured in **Hertz** (Hz).

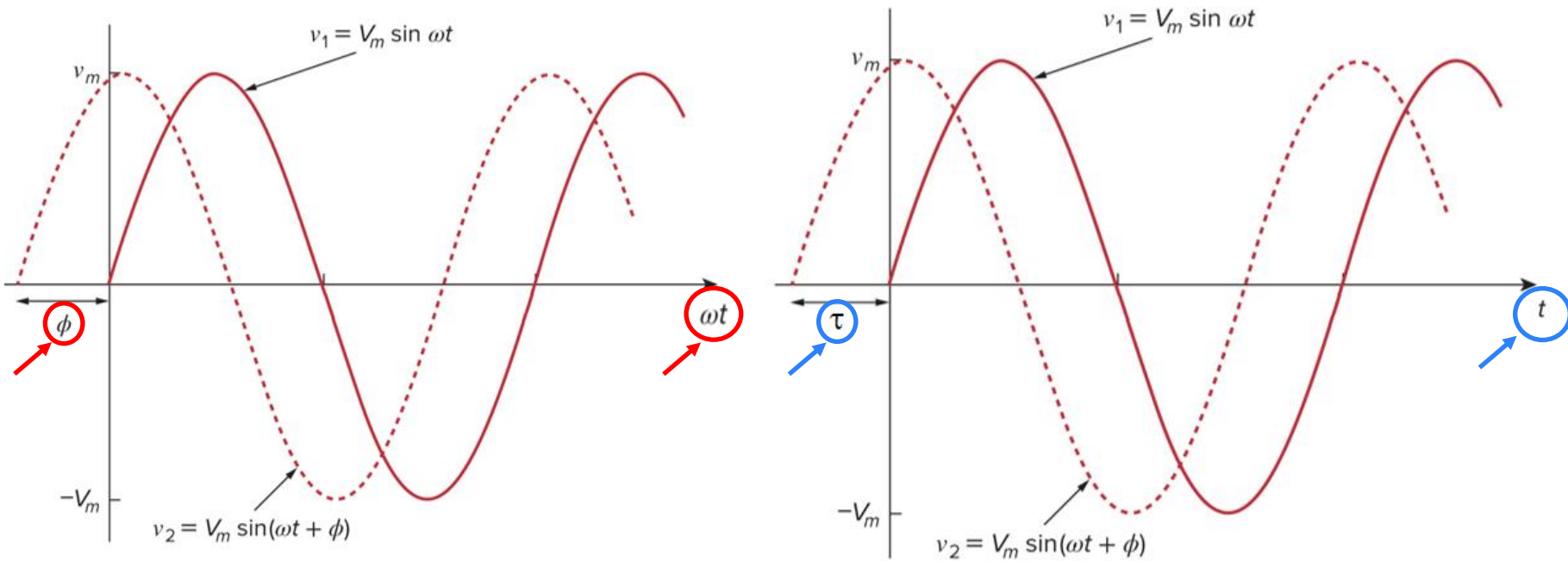
$$f = \frac{1}{T}$$

- Using $T = \frac{2\pi}{\omega}$, **angular frequency** is shown to be proportional to frequency:

$$\omega = 2\pi f$$

Sinusoids - Phase

- We can have sinusoids with different **phases**:
 $v_1(t) = V_m \sin(\omega t)$
 $v_2(t) = V_m \sin(\omega t + \phi)$



$$\phi = \frac{\tau}{T} \times 360^\circ$$

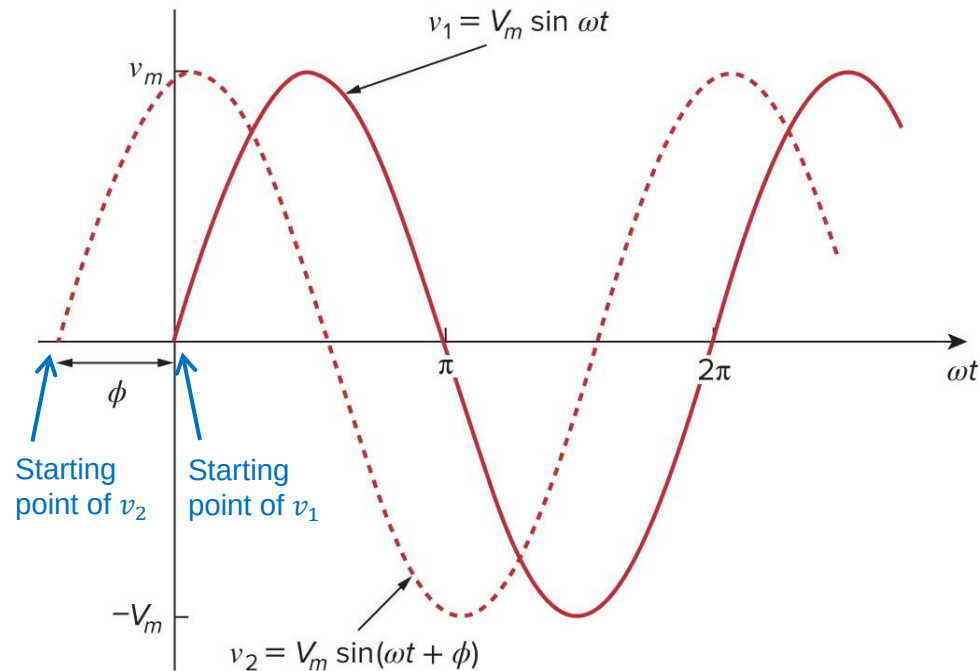
Sinusoids - Phase

- Given two sinusoids with different phases:

$$v_1(t) = V_m \sin(\omega t)$$

$$v_2(t) = V_m \sin(\omega t + \phi)$$

- If $\phi \neq 0$, v_1 and v_2 are **out of phase**.
- If $\phi = 0$, v_1 and v_2 are **in phase** (they reach their maxima and minima at exactly the same time).



- It can be assumed that v_1 starts at a **later** time ($t = 0$ s) compared to v_2 which begins **earlier** at $t = -\frac{\phi}{\omega}$ s. Thus,

v_1 is said to **lag** v_2
 v_2 is said to **lead** v_1

Note: We can compare them in this manner because they have the **same frequency**.

Sinusoids - Phase

- To evaluate a sinusoidal function at any value of time, it is required that **phase** is given in **radian** rather than in **degree** since **angular frequency** ω is given in **rad/s**.
- E.g. if you are asked to evaluate $v(t) = 50 \cos(30t + 10^\circ)$ for $t = 10$ ms you can do one of the following:

1. Convert radian to degree:

$$\phi^\circ = \phi \text{ rad} \times \frac{180}{\pi}$$

$$v(0.01) = 50 \cos(30 \times 0.01 \text{ rad} + 10^\circ)$$

$$\left(0.3 \times \frac{180}{\pi}\right) = 17.18^\circ$$

$\Rightarrow$

$$v(0.01) = 50 \cos(17.18^\circ + 10^\circ) = 44.47 \text{ V}$$

or

2. Convert degree to radian:

$$\phi \text{ rad} = \phi^\circ \times \frac{\pi}{180}$$

$$v(0.01) = 50 \cos(0.3 \text{ rad} + 10^\circ)$$

$$\left(10 \times \frac{\pi}{180}\right) = 0.174 \text{ rad}$$

$\Rightarrow$

$$v(0.01) = 50 \cos(0.3 \text{ rad} + 0.174 \text{ rad}) = 44.47 \text{ V}$$

Sinusoids – Sine vs cosine

- Sinusoids may be expressed as **sine** or **cosine** form.
- To **compare** two sinusoids, it is convenient to express both as either sine or cosine functions with **positive** amplitudes, using some **trigonometric identities**:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

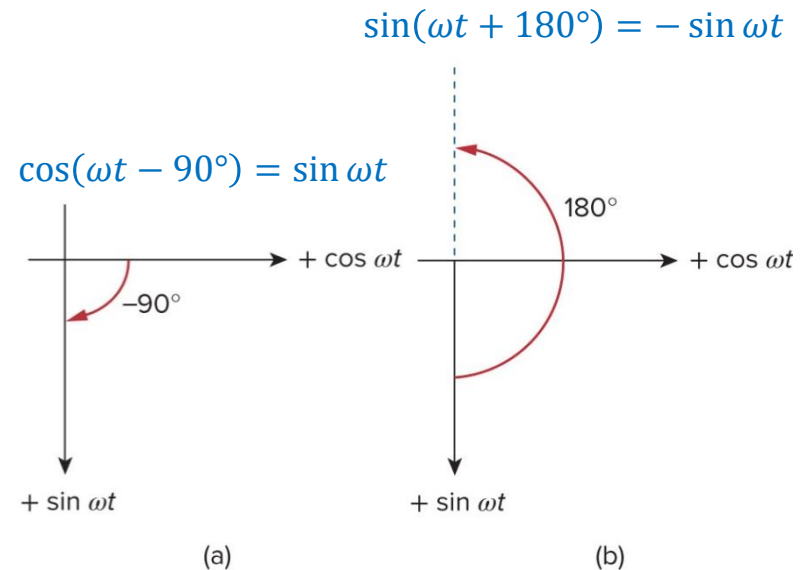
and

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$



Two examples of graphical representation for trigonometric identities (phase shift)

Exercise 1

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.



$$\begin{aligned}\cos(\omega t - 180^\circ) &= -\cos \omega t \\ \cos(\omega t - 90^\circ) &= \sin(\omega t)\end{aligned}$$

Exercise 1

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Express both sinusoids in **cosine** form with **positive** amplitude for comparison:

Use $\cos(\omega t - 180^\circ) = -\cos \omega t$ for v_1

$$v_1 = -10 \cos(\omega t + 50^\circ)$$

$$v_1 = 10 \cos(\omega t + 50^\circ - 180^\circ)$$

$$v_1 = 10 \cos(\omega t - 130^\circ)$$



$$\begin{cases} v_1 = 10 \cos(\omega t - 130^\circ) \\ v_2 = 12 \cos(\omega t - 100^\circ) \end{cases}$$

or

Use $\cos(\omega t - 90^\circ) = \sin(\omega t)$ for v_2 :

$$v_2 = 12 \sin(\omega t - 10^\circ)$$

$$v_2 = 12 \cos(\omega t - 10^\circ - 90^\circ)$$

$$v_2 = 12 \cos(\omega t - 100^\circ)$$

$$\begin{cases} v_1 = 10 \cos(\omega t + 230^\circ) \\ v_2 = 12 \cos(\omega t + 260^\circ) \end{cases}$$

It is clear that phase difference is 30° with v_2 **leading** v_1 by 30° .

Review of complex numbers

- A powerful method for representing sinusoids is the **phasor**.
- In order to understand how phasors work, we need to review some of the **complex numbers** properties first.
 - A complex number z can be represented in **rectangular** or **Cartesian** form as:

$$z = x + jy$$

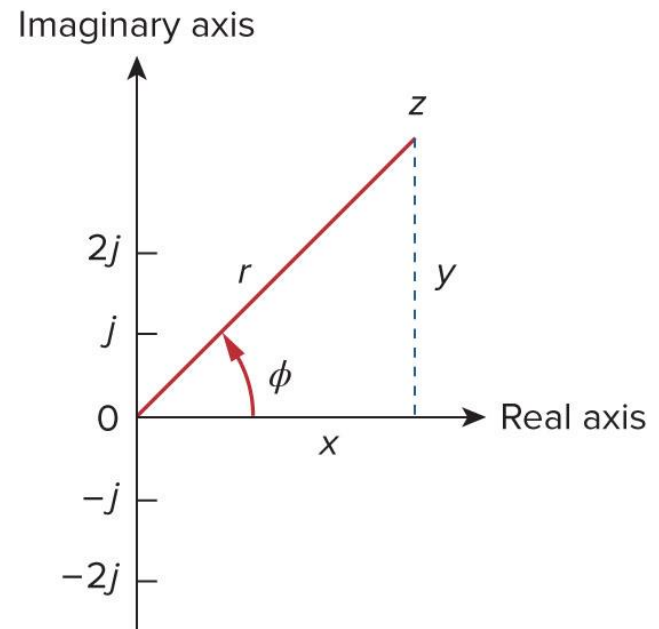
- It can also be written in **polar** or **exponential** form as:

Polar form:

$$z = r\angle\phi$$

Exponential form:

$$z = re^{j\phi}$$



- $j = \sqrt{-1}$: **Unit imaginary** number
- x : The **real** part of z
- y : The **imaginary** part of z
- r : The **magnitude** of z ($|z|$)
- ϕ : The **phase** of z

Review of complex numbers

- **Rectangular to polar** form transformation
when $z = x + jy$:

$$r = |z| = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1} \frac{y}{x}$$

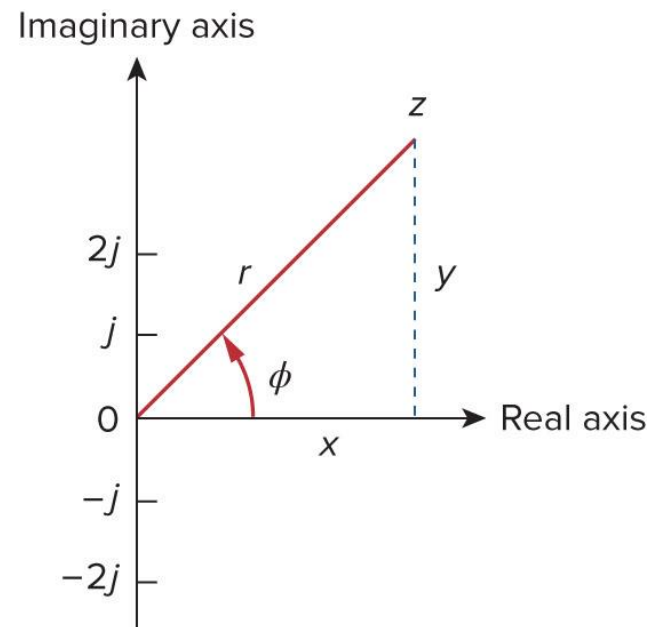
- **Polar to rectangular** form transformation
when $z = r \angle \phi$:

$$x = r \cos \phi$$

$$y = r \sin \phi$$



$$z = r \cos \phi + j r \sin \phi$$



Review of complex numbers

Given the complex numbers $z = x + jy = r\angle\phi = re^{j\phi}$,

$z_1 = x_1 + jy_1 = r_1\angle\phi_1 = r_1e^{j\phi_1}$, and $z_2 = x_2 + jy_2 = r_2\angle\phi_2 = r_2e^{j\phi_2}$, some basic properties to be used for phasor analysis are as follows:

- **Addition:**

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

- **Subtraction:**

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

- **Multiplication:**

$$z_1 z_2 = r_1 r_2 \angle(\phi_1 + \phi_2) = r_1 r_2 e^{j(\phi_1 + \phi_2)}$$

- **Division:**

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle(\phi_1 - \phi_2) = \frac{r_1}{r_2} e^{j(\phi_1 - \phi_2)}$$

Review of complex numbers

Given the complex numbers $z = x + jy = r\angle\phi = re^{j\phi}$,

$z_1 = x_1 + jy_1 = r_1\angle\phi_1 = r_1e^{j\phi_1}$, and $z_2 = x_2 + jy_2 = r_2\angle\phi_2 = r_2e^{j\phi_2}$, some basic properties to be used for phasor analysis are as follows:

- **Reciprocal:**

$$\frac{1}{z} = \frac{1}{r}\angle(-\phi) = \frac{1}{r}e^{-j\phi}$$

- **Square Root:**

$$\sqrt{z} = \sqrt{r}\angle\frac{\phi}{2} = \sqrt{r}e^{j\frac{\phi}{2}}$$

- **Complex Conjugate:**

$$z^* = x - jy = r\angle -\phi = re^{-j\phi}$$

$$\frac{1}{j} = -j$$

and

$$zz^* = r^2 = |z|^2$$

- **Also:** $e^{\pm j90^\circ} = \pm j$, $e^{\pm j180^\circ} = -1$

Phasor

- The idea of phasor representation is based on **Euler's identity**:

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

- We can write $\cos \phi$ and $\sin \phi$ as the **real part**, $\text{Re}(\cdot)$, and **imaginary part**, $\text{Im}(\cdot)$, of $e^{j\phi}$:

$$\begin{aligned}\cos \phi &= \text{Re}(e^{j\phi}) \\ \sin \phi &= \text{Im}(e^{j\phi})\end{aligned}$$

- Using the Euler's identity for a given sinusoid $v(t)$ we have:

$$v(t) = V_m \cos(\omega t + \phi) = \text{Re}(V_m e^{j(\omega t + \phi)})$$

$$v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$v(t) = \text{Re}(\mathbf{V} e^{j\omega t})$$



$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$$

$\mathbf{V} = V_m \angle \phi$ is the **phasor representation** of the sinusoid $v(t) = V_m \cos(\omega t + \phi)$

Phasor

- A **phasor** is a **complex number** that represents the **amplitude** and **phase** of a **sinusoid**.
 - For a given **sinusoid** in **cosine form**, $v(t) = V_m \cos(\omega t + \phi)$, the **phasor** is defined as a **complex number** by suppressing the time factor $e^{j\omega t}$, and taking the **amplitude** V_m and the **phase** ϕ .
 - To obtain the time-domain representation of a given phasor $\mathbf{V}$, use a **cosine function** with the **same magnitude** as the **phasor** and the **argument** ωt plus the **phase** of the phasor.

$$v(t) = V_m \cos(\omega t + \phi)$$

Time-domain representation



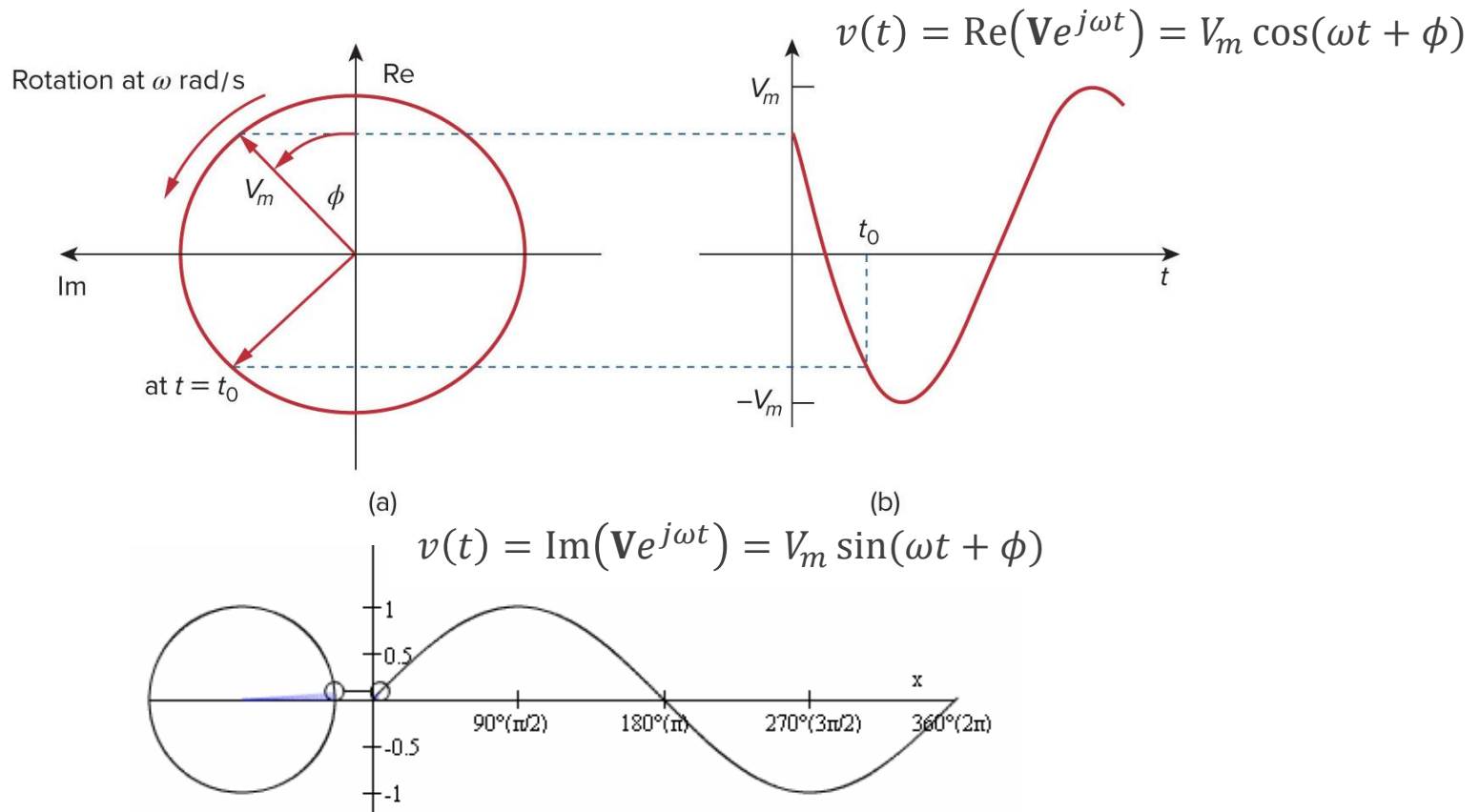
$$\mathbf{V} = V_m \angle \phi = V_m e^{j\phi}$$

Phasor-domain representation

- **Boldface** letters like $\mathbf{V}$ are used to represent phasors because they are **vector-like** quantities.

Phasor – Graphical representation

- Consider the plot of $\mathbf{v}e^{j\omega t} = V_m e^{j(\omega t + \phi)}$
 - As time increases, the vector rotates on a circle of **radius** V_m at an **angular velocity** ω in the **counter clockwise** direction.

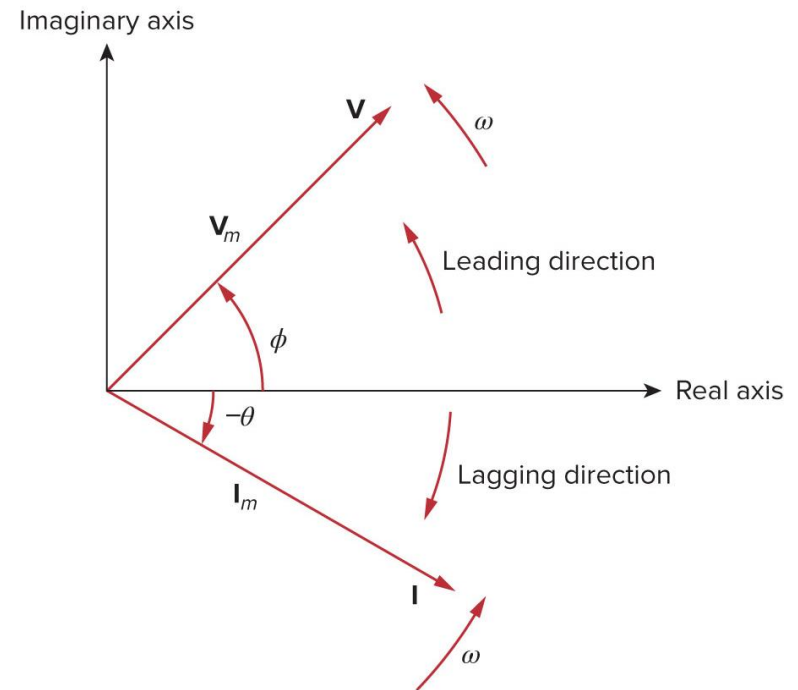


<https://giphy.com/gifs/mathematics-sin-pi-NKLdcqhwo2f8A/>

Phasor diagram

- A phasor can be expressed as a **vector** with a **magnitude** and a **phase** (direction).
- **Sketching** the phasors in a Cartesian coordinates or complex plane is called **phasor diagram**.
- Phasor diagram is plotted for **constant** and **equal** frequency ω , without being shown explicitly, that is why it is also known as **frequency domain**.

- The convention for measuring **angle/phase** is from **positive real axis** rotating **counter clockwise**.
- A **leading** phasor is the one **ahead** of other phasors in **counter clockwise** direction.



Phasor diagram showing
 $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$

Sinusoid-phasor transformation

- Phasors are defined in **cosine form** for both **voltage** and **current signals**.
- Any other form of sinusoid **should** be converted to cosine form using trigonometric identities.

Time-domain representation	Phasor-domain representation
$v(t) = V_m \cos(\omega t + \phi)$	$\mathbf{V} = V_m \angle \phi$
$v(t) = V_m \sin(\omega t + \phi)$	$\mathbf{V} = V_m \angle (\phi - 90^\circ)$
$v(t) = -V_m \cos(\omega t + \phi)$	$\mathbf{V} = V_m \angle (\phi \pm 180^\circ)$
$v(t) = -V_m \sin(\omega t + \phi)$	$\mathbf{V} = V_m \angle (\phi + 90^\circ)$
$i(t) = I_m \cos(\omega t + \theta)$	$\mathbf{I} = I_m \angle \theta$
$i(t) = I_m \sin(\omega t + \theta)$	$\mathbf{I} = I_m \angle (\theta - 90^\circ)$
$i(t) = -I_m \cos(\omega t + \theta)$	$\mathbf{I} = I_m \angle (\theta \pm 180^\circ)$
$i(t) = -I_m \sin(\omega t + \theta)$	$\mathbf{I} = I_m \angle (\theta + 90^\circ)$

Sinusoid-phasor transformation

- We can find the relationship between **linear mathematical operations** in **time domain** and their transformation in **phasor domain**.
- **Differentiating** a **sinusoid** is equivalent to **multiplying** its corresponding **phasor** by $j\omega$.
 - Consider a sinusoid and its corresponding phasor:

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \mathbf{V} = V_m \angle \phi$$

Take **time derivative** of the sinusoid:

$$\frac{dv}{dt} = -\omega V_m \sin(\omega t + \phi)$$

$$= \omega V_m \cos(\omega t + \phi + 90^\circ)$$

$$= \operatorname{Re}(\omega V_m e^{j(\omega t + \phi + 90^\circ)}) = \operatorname{Re}\left(\omega \underbrace{V_m e^{j\phi}}_{\mathbf{V}} e^{j\omega t} \underbrace{e^{j90^\circ}}_j\right) = \operatorname{Re}(j\omega \mathbf{V} e^{j\omega t})$$

$$\frac{dv}{dt}$$

(Time domain)



$$j\omega \mathbf{V}$$

(Phasor domain)

Sinusoid-phasor transformation

- **Integrating** a **sinusoid** is equivalent to **dividing** its corresponding **phasor** by $j\omega$.

- Consider the same sinusoid and its corresponding phasor:

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \mathbf{V} = V_m \angle \phi$$

Take **time integral** of the sinusoid:

$$\int v dt = \frac{1}{\omega} V_m \sin(\omega t + \phi)$$

$$= \frac{1}{\omega} V_m \cos(\omega t + \phi - 90^\circ)$$

$$= \operatorname{Re} \left(\frac{1}{\omega} V_m e^{j(\omega t + \phi - 90^\circ)} \right) = \operatorname{Re} \left(\frac{1}{\omega} \underbrace{V_m e^{j\phi}}_{\mathbf{V}} e^{j\omega t} \underbrace{e^{-j90^\circ}}_{-j = \frac{1}{j}} \right) = \operatorname{Re} \left(\frac{1}{j\omega} \mathbf{V} e^{j\omega t} \right)$$

$$\int v dt$$

(Time domain)



$$\frac{1}{j\omega} \mathbf{V}$$

(Phasor domain)

Some notes on sinusoids and phasors

- There are some conceptual differences between sinusoids and their corresponding phasors.

$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m \angle \phi$$

1. $v(t)$ is the **instantaneous** or **time-domain** representation of a sinusoid, while $\mathbf{V}$ is the **phasor** or **frequency-domain** representation of the sinusoid.
2. $v(t)$ is **time dependent**, while $\mathbf{V}$ is **constant**.
3. $v(t)$ is always a **real function** with **no** complex term, while $\mathbf{V}$ is generally a **complex number**.
4. Keep in mind that **phasor analysis** on one or multiple sinusoids applies **only** when the **frequency is constant** and they are of the **same frequency**.
 - **Note:** Phasor analysis for a circuit with **multiple frequencies** is possible but requires the use of **superposition**, and at **each stage** you have a **different set of phasors**.

Exercise 2

Transform these sinusoids to phasors:

a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

c) $i_1 = -8 \cos(16t + 15^\circ) \text{ A}$



$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \mathbf{V} = V_m \angle \phi$$

Exercise 2

Transform these sinusoids to phasors:

a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

c) $i_1 = -8 \cos(16t + 15^\circ) \text{ A}$

a) $\mathbf{I} = 6 \angle -40^\circ \text{ A}$

b) Convert to positive cosine form:

$$v = 4 \cos(30t + 50^\circ + 90^\circ) \text{ V} \quad \Rightarrow \quad \mathbf{V} = 4 \angle 140^\circ \text{ V}$$

c) Convert to positive cosine form (maintain the phase in the range of $\pm 180^\circ$):

$$i_1 = 8 \cos(16t + 15^\circ - 180^\circ) \text{ A} \quad \Rightarrow \quad \mathbf{I}_1 = 8 \angle -165^\circ \text{ A}$$



$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m \angle \phi$$

Exercise 3

- Find the sinusoids represented by these phasors:

a) $\mathbf{I} = -3 + j4 \text{ A}$

b) $\mathbf{V} = j8e^{-j20^\circ} \text{ V}$

c) $\mathbf{V}_1 = -25\angle 40^\circ \text{ V}$



Cartesian to polar transformation: $z = \sqrt{x^2 + y^2} \angle \tan^{-1}(y/x)$

Exercise 3

- Find the sinusoids represented by these phasors:

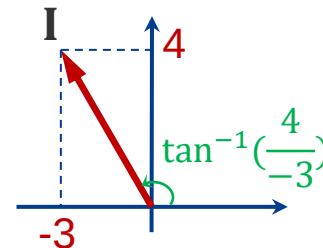
a) $\mathbf{I} = -3 + j4 \text{ A}$

b) $\mathbf{V} = j8e^{-j20^\circ} \text{ V}$

c) $\mathbf{V}_1 = -25\angle 40^\circ \text{ V}$

a) $\mathbf{I} = -3 + j4 = \sqrt{3^2 + 4^2} \angle \tan^{-1}\left(\frac{4}{-3}\right) = 5\angle 126.87^\circ \text{ A}$

Thus: $i = 5 \cos(\omega t + 126.87^\circ) \text{ A}$



b) Recall that $j = 1e^{j90^\circ} = 1\angle 90^\circ$:

$$\mathbf{V} = j8\angle(-20^\circ) = (1\angle 90^\circ) \times (8\angle(-20^\circ)) = 8\angle(90^\circ - 20^\circ) = 8\angle(70^\circ) \text{ V}$$

Thus: $v = 8 \cos(\omega t + 70^\circ) \text{ V}$

c) Convert to positive magnitude (maintain the phase in the range of $\pm 180^\circ$):

$$\mathbf{V}_1 = -25\angle 40^\circ = 25\angle(40^\circ - 180^\circ) = 25\angle(-140^\circ)$$

Thus: $v_1 = 25 \cos(\omega t - 140^\circ) \text{ V}$



Cartesian to polar transformation: $z = \sqrt{x^2 + y^2} \angle \tan^{-1}(y/x)$

Exercise 4

Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find $i_1(t) + i_2(t)$.



Polar to Cartesian transformation $z = r \cos \phi + j r \sin \phi$

Cartesian to Polar transformation: $z = \sqrt{x^2 + y^2} \angle \tan^{-1}(y/x)$

Exercise 4

Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find $i_1(t) + i_2(t)$.

We can use phasor analysis for summing sinusoid with the same frequency.

The phasor for current i_1 : $\mathbf{I}_1 = 4\angle 30^\circ$ A

The phasor for current i_2 is obtained by converting it to cosine form :

$$i_2 = 5 \cos(\omega t - 20^\circ - 90^\circ) \text{ A} \Rightarrow \mathbf{I}_2 = 5\angle -110^\circ \text{ A}$$

$$\text{Therefore, } i_1(t) + i_2(t) \Leftrightarrow \mathbf{I}_1 + \mathbf{I}_2 = 4\angle 30^\circ + 5\angle -110^\circ$$

Use Polar to Cartesian transformation $z = r \cos \phi + j r \sin \phi$

$$\begin{aligned} \mathbf{I}_1 + \mathbf{I}_2 &= (4 \cos 30^\circ + j4 \sin 30^\circ) + (5 \cos(-110^\circ) + j \sin(-110^\circ)) \\ &= 3.464 + j2 - 1.71 - j4.698 = 1.754 - j2.698 \end{aligned}$$

Use Cartesian to Polar transformation: $z = \sqrt{x^2 + y^2} \angle \tan^{-1}(y/x)$

$$\mathbf{I}_1 + \mathbf{I}_2 = \sqrt{1.754^2 + 2.698^2} \angle \tan^{-1}\left(\frac{-2.698}{1.754}\right) = 3.218 \angle -56.97^\circ \text{ A}$$

$$\text{Therefore, } i_1(t) + i_2(t) = 3.218 \cos(\omega t - 56.97^\circ) \text{ A}$$

Exercise 5

If $v_1(t) = -10 \sin(\omega t - 30^\circ)$ V and $v_2(t) = 20 \cos(\omega t + 45^\circ)$ V, find $v(t) = v_1(t) + v_2(t)$

- For practice!
- Answer: $v(t) = 29.77 \cos(\omega t + 49.98^\circ)$

Phasor relationships for resistor

- Current and voltage relationships can be **mapped** from **time domain** into **phasor domain** very simply for passive elements like resistors, capacitors and inductors.
- For a resistor R , if the **current** is given as a **sinusoid** $i = I_m \cos(\omega t + \phi)$ (AC) with the corresponding current phasor of $\mathbf{I} = I_m \angle \phi$, the voltage is obtained using **Ohm's law** as below:

$$v = Ri = RI_m \cos(\omega t + \phi)$$

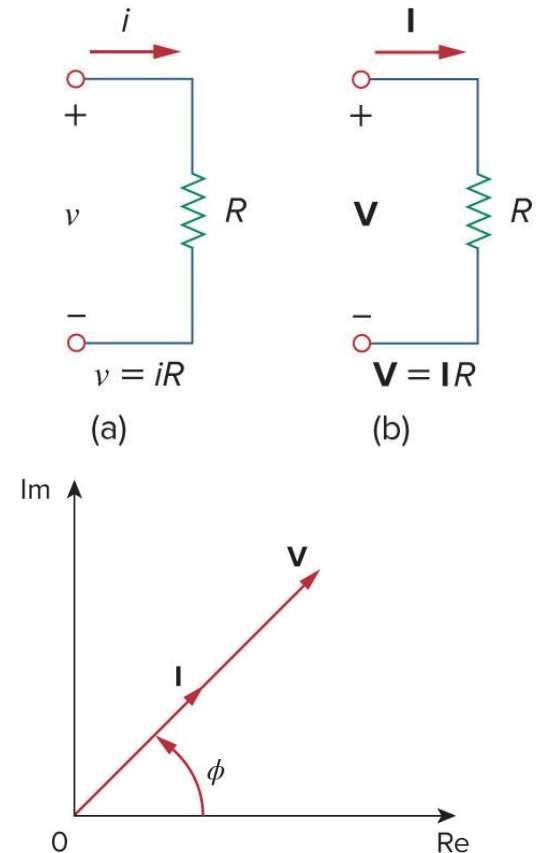
The phasor representation of the voltage is:

$$\mathbf{V} = RI_m \angle \phi = R\mathbf{I}$$

Therefore:

$$\mathbf{V} = R\mathbf{I}$$

Voltage and **current** of a **resistor** are **in phase** and related via **Ohm's law** in **phasor domain**.



Phasor relationships for inductor

- For an inductor L , if the **current** is given as a **sinusoid** $i = I_m \cos(\omega t + \phi)$ (AC) with the corresponding current phasor of $\mathbf{I} = I_m \angle \phi$, the voltage is obtained as below:

$$v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi)$$

Recall that $-\sin a = \cos(a + 90^\circ)$:

$$v = \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

The voltage phasor is: $\mathbf{V} = \omega L I_m \angle (\phi + 90^\circ)$

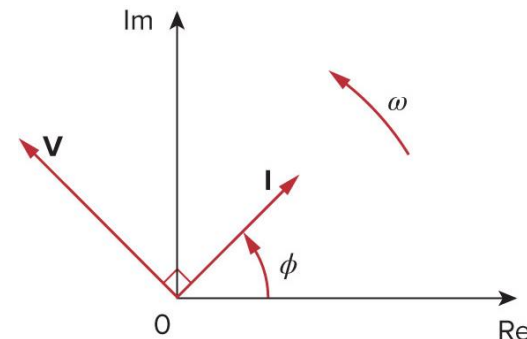
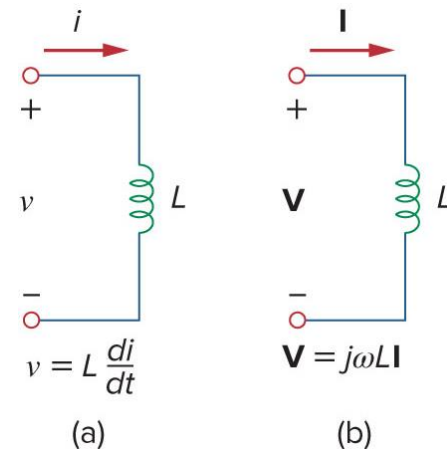
Recall that $1 \angle 90^\circ = e^{j90^\circ} = j$:

$$\mathbf{V} = \omega L I_m e^{j\phi} e^{j90^\circ} = j\omega L I_m \angle \phi = j\omega L \mathbf{I}$$

Therefore:

$$\mathbf{V} = j\omega L \mathbf{I}$$

- Voltage** and **current** of an inductor are **90° out of phase**.
- The **inductor current** “lags” its **voltage** by 90°.



Phasor relationships for capacitor

- For a capacitor C , if the **voltage** is given as a **sinusoid** $v = V_m \cos(\omega t + \phi)$ (AC) with the corresponding voltage phasor of $\mathbf{V} = V_m \angle \phi$, the current is obtained as below:

$$i = C \frac{dv}{dt} = -\omega C V_m \sin(\omega t + \phi)$$

$$i = \omega C V_m \cos(\omega t + \phi + 90^\circ)$$

The current phasor is:

$$\mathbf{I} = \omega C V_m \angle (\phi + 90^\circ)$$

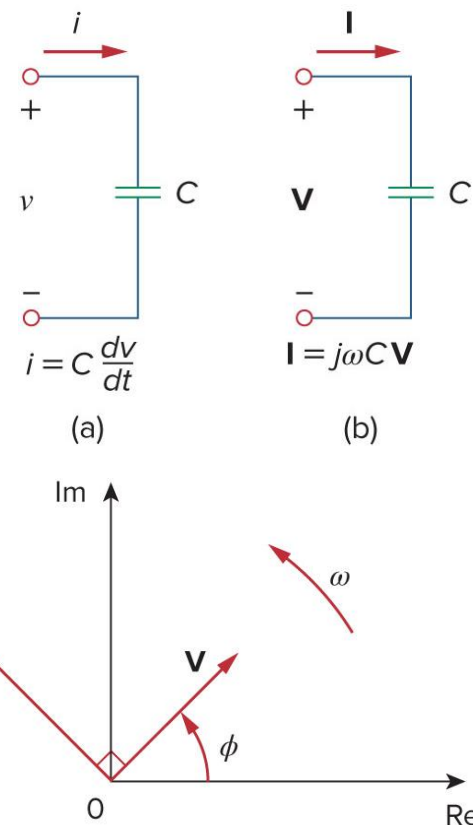
$$\mathbf{I} = \omega C V_m e^{j\phi} e^{j90^\circ} = j\omega C V_m \angle \phi = j\omega C \mathbf{V}$$

$$\mathbf{I} = j\omega C \mathbf{V}$$

Therefore:

$$\mathbf{V} = \frac{1}{j\omega C} \mathbf{I}$$

- Voltage** and **current** of a **capacitor** are **90° out of phase**.
- The **capacitor current** “leads” its **voltage** by 90°.



Passive elements phasors

- Summary of time-domain and phasor-domain representations of the passive circuit elements:

Element	Time-domain representation	Phasor-domain representation
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C \frac{dv}{dt}$	$\mathbf{V} = \frac{1}{j\omega C} \mathbf{I}$

Impedance

- **Ohm's law** can be extended to capacitors and inductors in **phasor domain**.
- The **impedance** Z of a circuit element is the **ratio** of the **phasor voltage** V to the **phasor current** I , measured in **ohms** (Ω).

Recall the **voltage-current** relations for the three passive elements in **phasor domain**:

$$V = RI$$

Resistor

$$V = j\omega LI$$

Inductor

$$V = \frac{1}{j\omega C} I$$

Capacitor

Write these expression in terms of the ratio of the phasor voltage to the phasor current:

$$\frac{V}{I} = R$$

Resistor

$$\frac{V}{I} = j\omega L$$

Inductor

$$\frac{V}{I} = \frac{1}{j\omega C}$$

Capacitor

- These **impedances** represent **Ohm's law** in the **phasor domain**:

$$Z = \frac{V}{I}$$

or

$$V = ZI$$

Impedance

- Given the impedances of resistors, inductors, and capacitors, only the impedance of the **resistor** is a **constant real value**.
- The impedance of the **inductor** and **capacitor** are **frequency-dependent** complex values.

$$Z_R = R$$

$$Z_L = j\omega L$$

$$Z_C = \frac{1}{j\omega C}$$

- Consider two **extreme cases** for **angular frequency** ω :

1. At **low** frequencies: $\omega = 0$

$\Rightarrow$

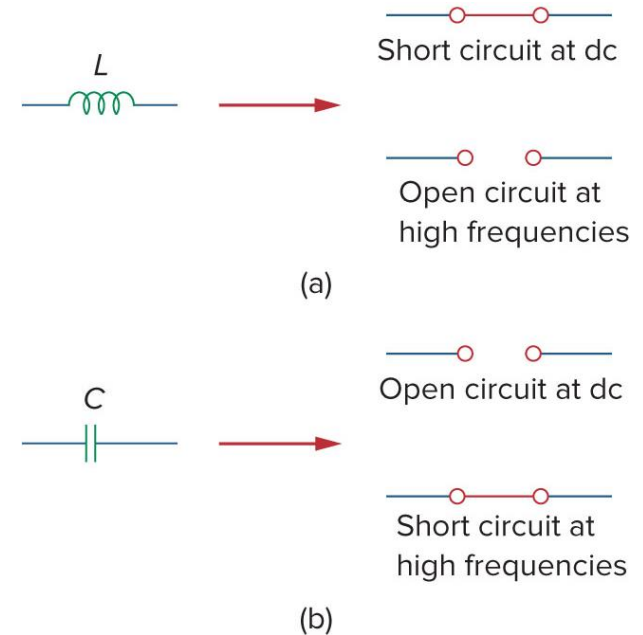
$$\begin{aligned} Z_L &= 0 \\ Z_C &\rightarrow \infty \end{aligned}$$

2. At **high** frequencies: $\omega \rightarrow \infty$

$\Rightarrow$

$$\begin{aligned} Z_L &\rightarrow \infty \\ Z_C &= 0 \end{aligned}$$

- At **low frequencies**, **inductor** acts like a **short circuit** while **capacitor** acts like an **open circuit**.
- At **high frequencies**, **Inductor** acts like an **open circuit** while **capacitor** acts like a **short circuit**.



Impedance

- The **impedances** of resistors, inductors, and capacitors **can be combined** together as one impedance.
- **Impedance**, as a **complex quantity**, can be expressed in **rectangular** or **polar** forms.
 - $R = \text{Re}(\mathbf{Z})$: **Resistance**.
 - $X = \text{Im}(\mathbf{Z})$: **Reactance**.
 - The impedance $\mathbf{Z}$, resistance R , and reactance X , are all measured in ohms (Ω).

$$\mathbf{Z} = R + jX$$



$$\mathbf{Z} = |\mathbf{Z}| \angle \theta$$

$P \rightarrow R$

$$R = |\mathbf{Z}| \cos \theta$$

$$X = |\mathbf{Z}| \sin \theta$$

$R \rightarrow P$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}$$

$$\theta = \tan^{-1} \left(\frac{X}{R} \right)$$

Impedance

- Inductors and capacitors have **pure imaginary** impedances:

$$Z_L = j\omega L$$

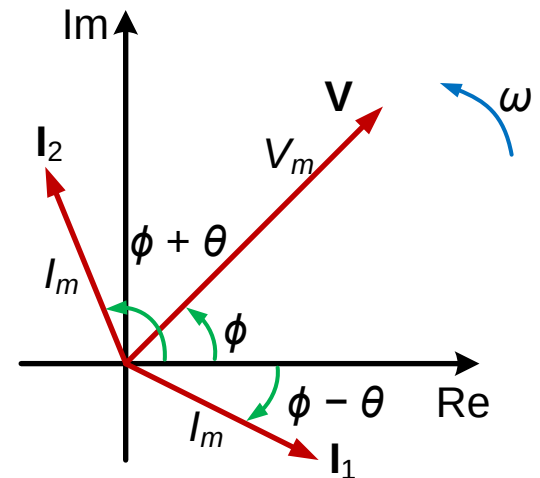
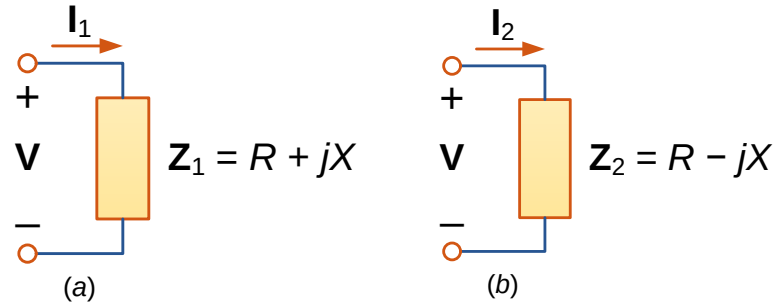
$$Z_C = \frac{1}{j\omega C} = -j \frac{1}{\omega C}$$

- Also, an **impedance** in general is expressed as a **complex number** with either **positive** or **negative** imaginary part:

$$Z = R \pm jX$$

$Z = R + jX$: **Inductive** or **lagging**
(current **lags** voltage, I_1).

$Z = R - jX$: **Capacitive** or **leading**
(current **leads** voltage, I_2).



Impedance

Proof: Consider the following circuits with equal input voltage phasor $\mathbf{V} = V_m \angle \phi$, and complex conjugate impedances $\mathbf{Z}_1 = R + jX$ and $\mathbf{Z}_2 = R - jX$ ($|\mathbf{Z}_1| = |\mathbf{Z}_2|$).

Based on Ohm's law:

$$\mathbf{I}_1 = \frac{\mathbf{V}}{\mathbf{Z}_1} = \frac{V_m \angle \phi}{R + jX} = \frac{V_m \angle \phi}{\sqrt{R^2 + X^2} \angle \tan^{-1} \left(\frac{X}{R} \right)}$$

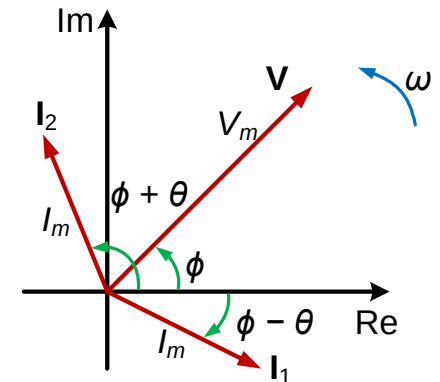
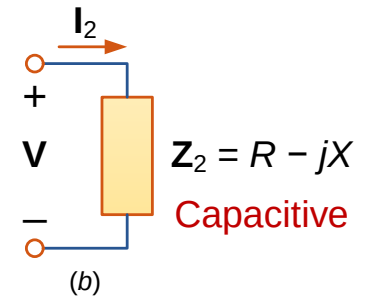
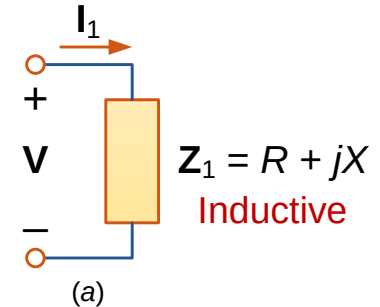
$$\mathbf{I}_1 = \frac{V_m}{\underbrace{\sqrt{R^2 + X^2}}_{|\mathbf{Z}_1|}} \angle \left(\phi - \underbrace{\tan^{-1} \left(\frac{X}{R} \right)}_{\theta} \right) = \frac{V_m}{\underbrace{|\mathbf{Z}_1|}_{I_m}} \angle (\phi - \theta) = I_m \angle (\phi - \theta)$$

and:

$$\mathbf{I}_2 = \frac{\mathbf{V}}{\mathbf{Z}_2} = \frac{V_m \angle \phi}{R - jX} = \frac{V_m \angle \phi}{\sqrt{R^2 + X^2} \angle \tan^{-1} \left(\frac{-X}{R} \right)}$$

$\tan^{-1}(-a) = -\tan^{-1}(a)$
odd function

$$\mathbf{I}_2 = \frac{V_m}{\underbrace{\sqrt{R^2 + X^2}}_{|\mathbf{Z}_2| = |\mathbf{Z}_1|}} \angle \left(\phi + \underbrace{\tan^{-1} \left(\frac{X}{R} \right)}_{\theta} \right) = \frac{V_m}{\underbrace{|\mathbf{Z}_1|}_{I_m}} \angle (\phi + \theta) = I_m \angle (\phi + \theta)$$



Admittance

- **Impedance** represents the **resistance** of a circuit element to the **flow** of **sinusoidal current**.
 - Impedance is not a phasor, it is just a complex number.
- **Admittance** is simply the **reciprocal** of the impedance. It is measured in **siemens (S)**.

$$Y = \frac{1}{Z}$$

or

$$Y = \frac{I}{V}$$

- The impedance and admittance of resistors, inductors, and capacitors can be obtained as follows:

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

Admittance

- Admittance $\mathbf{Y}$ is also a **complex quantity**.
 - $G = \text{Re}(\mathbf{Y})$: **Conductance**.
 - $B = \text{Im}(\mathbf{Y})$: **Susceptance**.
 - Admittance Y , conductance G , and susceptance B , are all measured in **siemens** (S).

$$\mathbf{Y} = G + jB$$

- The admittance $\mathbf{Y} = G + jB$ and impedance $\mathbf{Z} = R + jX$ components are related:

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} \quad \rightarrow \quad G + jB = \frac{1}{R + jX}$$

To find the relationship, cancel j in the denominator (i.e. multiply the fraction by the **complex conjugate** of $R + jX$, since $\mathbf{Z}\mathbf{Z}^* = |\mathbf{Z}|^2$).

$$G + jB = \frac{1}{R + jX} \times \frac{R - jX}{R - jX} = \frac{R - jX}{R^2 + X^2} \quad \rightarrow$$

$$G = \frac{R}{R^2 + X^2}$$

$$B = \frac{-X}{R^2 + X^2}$$

- Similarly, $\mathbf{Z}$ components are related to $\mathbf{Y}$:

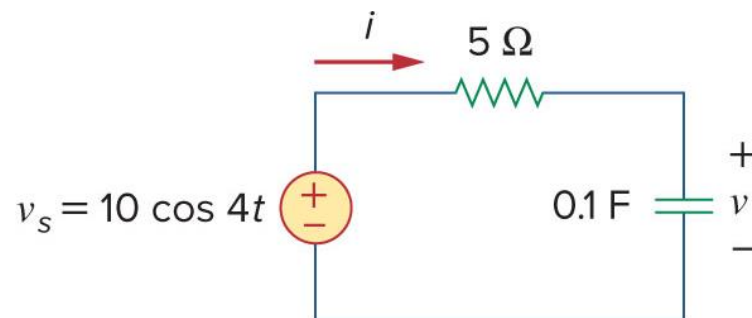
$$R = \frac{G}{G^2 + B^2}$$

$$X = \frac{-B}{G^2 + B^2}$$

Exercise 6

Find $v(t)$ and $i(t)$ in the circuit below.

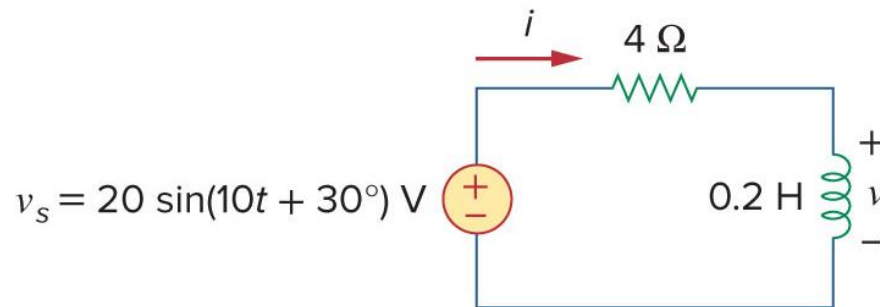
Solution in Exercises_solution_Topic_7.pdf



Exercise 7

Find $v(t)$ and $i(t)$ in the circuit below

- For practice!
- Answer: $v(t) = 8.944 \cos(10t + 3.43^\circ) \text{ V}$
 $i(t) = 4.472 \cos(10t - 86.57^\circ) \text{ A}$



Kirchhoff's laws

- **KVL** and **KCL** applies to **sinusoidal AC circuits** in the **same way** as in DC circuits by using the **sum of voltage phasors** for **KVL** and the **sum of current phasors** for **KCL**.
- For a single loop circuit with **voltages** $v_1, v_2, \dots, v_n$, we can apply **KVL**:

$$v_1 + v_2 + \dots + v_n = 0$$

- Assuming sinusoidal steady-state conditions, all voltages in the circuit can be written in cosine form:

$$V_{m1} \cos(\omega t + \phi_1) + V_{m2} \cos(\omega t + \phi_2) + \dots + V_{mn} \cos(\omega t + \phi_n) = 0$$

- For each sinusoid, we can use the corresponding phasor in the sum:

$$V_{mk} \cos(\omega t + \phi_k) \Leftrightarrow \mathbf{V}_k = V_{mk} \angle \phi_k \text{ for } k = 1, 2, \dots, n$$

Therefore:

$$\mathbf{V}_1 + \mathbf{V}_2 + \dots + \mathbf{V}_n = 0$$

- Using the same procedure for **currents** in a single node and applying **KCL** using the current phasors:

$$I_{mk} \cos(\omega t + \theta_k) \Leftrightarrow \mathbf{I}_k = I_{mk} \angle \theta_k \text{ for } k = 1, 2, \dots, n \rightarrow \mathbf{I}_1 + \mathbf{I}_2 + \dots + \mathbf{I}_n = 0$$

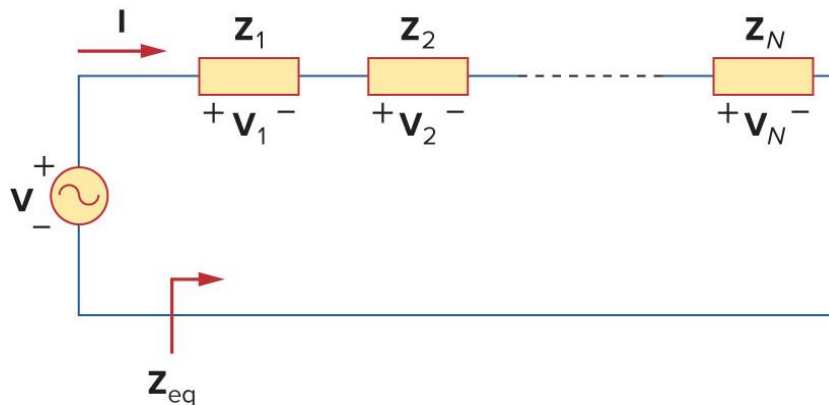
Impedances in series

- The **equivalent impedance** Z_{eq} of **series-connected** impedances is the **sum** of the **individual** impedances (**same** as **series** connection of **resistors**).
 - Consider N **series-connected impedances** as shown in the circuit below.
 - The **same** current I flows through the impedances.
 - Apply KVL in phasor domain:

$$V = V_1 + V_2 + \cdots + V_n = I(Z_1 + Z_2 + \cdots + Z_n)$$

- The **equivalent impedance** Z_{eq} at the input terminals is:

$$Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \cdots + Z_n$$



$$Z_{eq} = Z_1 + Z_2 + \cdots + Z_n$$

Voltage division

- **Voltage division** for **impedances** works the **same** way as voltage division for **resistors**.

- For **two impedances** connected **in series** as shown in the circuit, the current **I** is obtained using Ohm's law:

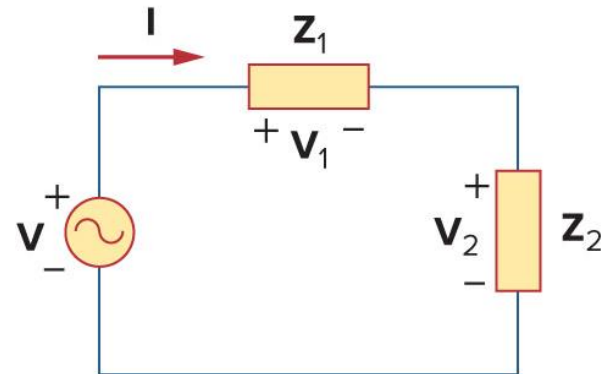
$$I = \frac{V}{Z_1 + Z_2}$$

- Also, $V_1 = Z_1 I$ and $V_2 = Z_2 I$. Therefore, each voltage is obtained as follows:

Voltage
Division
Principle

$$V_1 = \frac{Z_1}{Z_1 + Z_2} V$$

$$V_2 = \frac{Z_2}{Z_1 + Z_2} V$$



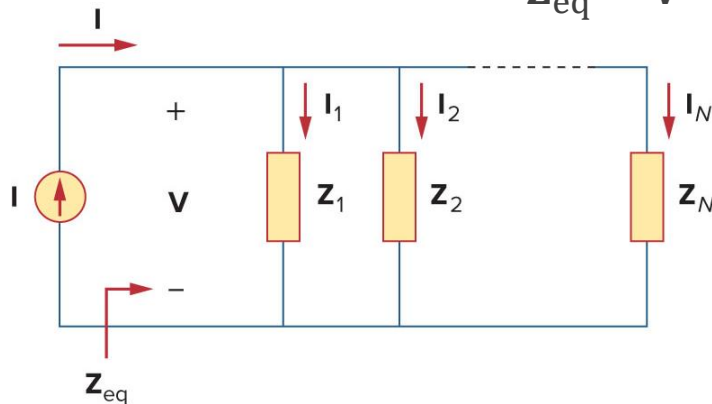
Impedances in parallel

- The **equivalent impedance** Z_{eq} of **parallel-connected** impedances is the **reciprocal** of the **sum** of the **individual reciprocal** impedances (**same** as **parallel** connection of **resistors**).
 - Consider N **parallel-connected impedances** as shown in the circuit below.
 - The **voltage** across each impedance is **same**.
 - Apply KCL in phasor domain at the top node:

$$I = I_1 + I_2 + \dots + I_n = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n} \right)$$

- The **equivalent impedance** Z_{eq} at the input terminals is:

$$\frac{1}{Z_{eq}} = \frac{I}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n}$$



$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n}$$

OR (using **admittance**):

$$Y_{eq} = Y_1 + Y_2 + \dots + Y_n$$

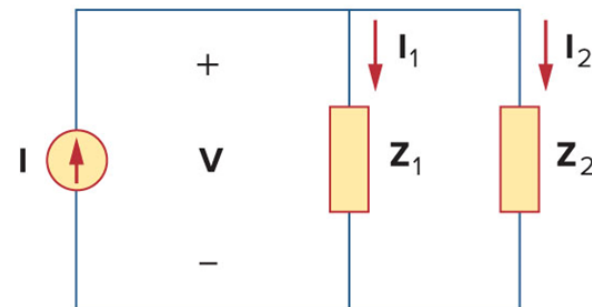
Current division

- **Current division** for **impedances** works the **same** way as current division for **resistors**.
 - For **two impedances** connected **in parallel** as shown in the circuit, the equivalent impedance $\mathbf{Z}_{eq}$ is obtained as:
$$\mathbf{Z}_{eq} = \frac{1}{\frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2}} = \frac{\mathbf{Z}_1 \mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$$
 - Also, $\mathbf{V} = \mathbf{Z}_{eq} \mathbf{I} = \mathbf{Z}_1 \mathbf{I}_1 = \mathbf{Z}_2 \mathbf{I}_2$. Therefore, each current is obtained as follows:

Current
Division
Principle

$$\mathbf{I}_1 = \frac{\mathbf{Z}_1 \parallel \mathbf{Z}_2}{\mathbf{Z}_1} \mathbf{I} = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{I}$$

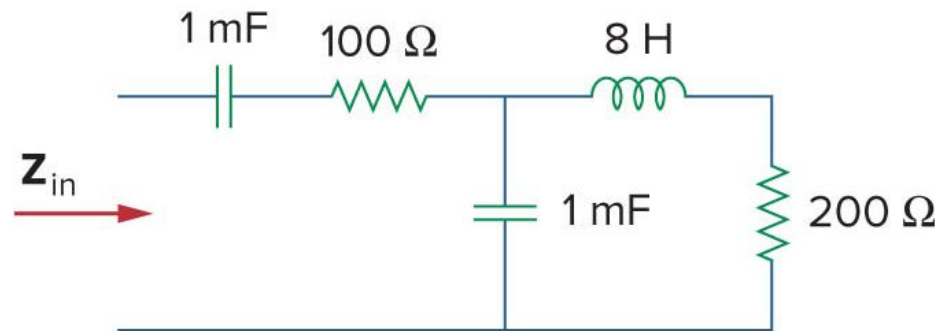
$$\mathbf{I}_2 = \frac{\mathbf{Z}_1 \parallel \mathbf{Z}_2}{\mathbf{Z}_2} \mathbf{I} = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{I}$$



Exercise 8

Determine the input impedance of the circuit shown below assuming the circuit operates at $f = \frac{20}{2\pi}$ Hz.

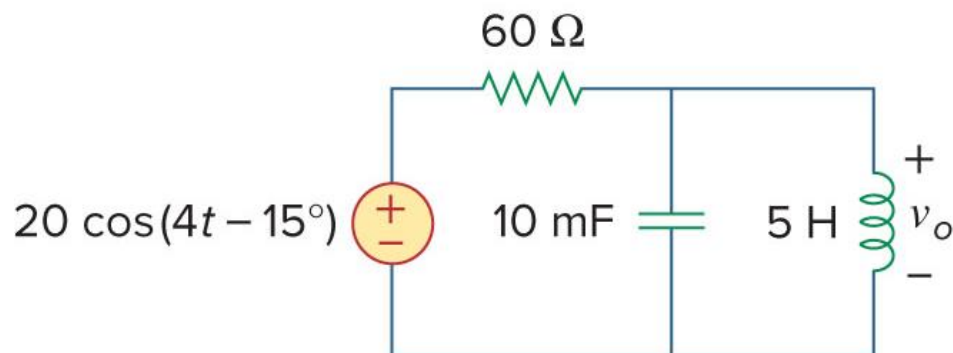
Solution in Exercises_solution_Topic_7.pdf



Exercise 9

Find the output voltage $v_o(t)$ in the circuit below.

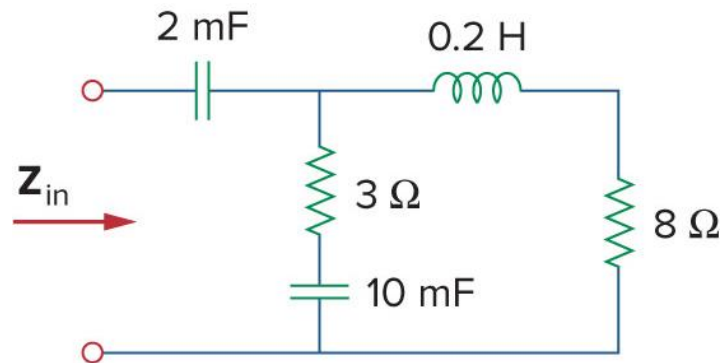
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Exercise 10

Find the input impedance of the circuit shown below assuming the circuit operates at $\omega = 50 \text{ rad/s}$

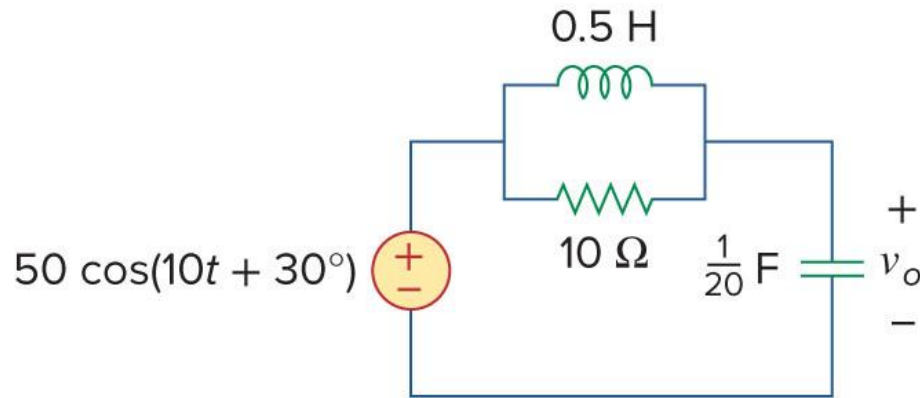
- For practice!
- Answer: $\mathbf{Z}_{\text{in}} = 3.22 - j11.07 \, \Omega$



Exercise 11

Calculate the output voltage $v_o(t)$ in the circuit below.

- For practice!
- Answer: $v_o(t) = 35.36 \cos(10t - 105^\circ) \text{ V}$



Questions?

